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MASTER 1: Réseaux et Systèmes Informatique

Rapport d'Installation et de Configuration de Graylog

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Introduction

Ce rapport décrit les étapes d'installation et de configuration de la solution de journalisation centralisée Graylog sur un système Ubuntu. L'objectif est de mettre en place une plateforme performante pour la collecte, le traitement et l'analyse des logs système.

Prérequis

- Système: Ubuntu 24.04 LTS
- Accès root ou sudo
- Ports ouverts: 9000 (Graylog), 9200 (Elasticsearch), 27017 (MongoDB)
- Mises à jour:

```
root@ababacar-Graylog:/home/ababacar# apt update -y
Hit:1 http://sn.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://sn.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://sn.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@ababacar-Graylog:/home/ababacar# apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@ababacar-Graylog:/home/ababacar#
```

Installation des composants

```
root@ababacar-Graylog:/home/ababacar# apt install apt-transport-https gnupg2 uuid-runtime pwgen curl dirmngr -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
pwgen is already the newest version (2.08-2build1).
curl is already the newest version (7.81.0-1ubuntu1.20).
dirmngr is already the newest version (2.2.27-3ubuntu2.3).
uuid-runtime is already the newest version (2.37.2-4ubuntu3.4).
apt-transport-https is already the newest version (2.4.14).
gnupg2 is already the newest version (2.2.27-3ubuntu2.3).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@ababacar-Graylog:/home/ababacar#
```

Installation d'un JDK

```
root@ababacar-Graylog:/home/ababacar# apt install openjdk-11-jre-headless -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-11-jre-headless is already the newest version (11.0.27+6~us1-0ubuntu1~22.04).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@ababacar-Graylog:/home/ababacar#
```

Verification de la version de Java

```
root@ababacar-Graylog:/home/ababacar# java -version
openjdk version "11.0.27" 2025-04-15
OpenJDK Runtime Environment (build 11.0.27+6-post-Ubuntu-0ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.27+6-post-Ubuntu-0ubuntu122.04, mixed mode, sharing)
root@ababacar-Graylog:/home/ababacar#
```

Mise a jour et installation

```
root@ababacar-Graylog:/home/ababacar# sudo apt update && sudo apt install -y apt-transport-https openjdk-17-jre-headless uid-runtime
pwgen gnupg curl wget
Hit:1 http://sn.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://sn.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://sn.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 https://artifacts.elastic.co/packages/oss-7.x/apt stable InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
pwgen is already the newest version (2.08-2build1).
curl is already the newest version (7.81.0-1ubuntu1.20).
gnupg is already the newest version (2.2.27-3ubuntu2.3).
uid-runtime is already the newest version (2.37.2-4ubuntu3.4).
wget is already the newest version (1.21.2-2ubuntu1.1).
apt-transport-https is already the newest version (2.4.14).
openjdk-17-jre-headless is already the newest version (17.0.15+6-us1-0ubuntu1~22.04).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@ababacar-Graylog:/home/ababacar#
```

MongoDB

```
root@ababacar-Graylog:/home/ababacar# wget -qO - https://www.mongodb.org/static/pgp/server-6.0.asc | sudo gpg --dearmor -o /usr/share
/keyrings/mongodb-server-6.0.gpg
echo "deb [signed-by=/usr/share/keyrings/mongodb-server-6.0.gpg] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 multiverse
" | sudo tee /etc/apt/sources.list.d/mongodb-org-6.0.list

sudo apt update
sudo apt install -y mongodb-org

sudo systemctl enable mongod
sudo systemctl start mongod
deb [signed-by=/usr/share/keyrings/mongodb-server-6.0.gpg] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 multiverse
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://sn.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://sn.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 https://artifacts.elastic.co/packages/oss-7.x/apt stable InRelease
Hit:5 http://sn.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:6 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 InRelease [4 009 B]
Get:7 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0/multiverse amd64 Packages [99,4 kB]
```

OpenSearch (ou Elasticsearch)

```

root@ababacar-Graylog:/home/ababacar# wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor | sudo tee /usr/share/keyrings/elasticsearch-keyring.gpg > /dev/null
echo "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/oss-7.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elastic-7.x.list

sudo apt update
sudo apt install -y elasticsearch-oss

sudo systemctl daemon-reexec
sudo systemctl enable elasticsearch
sudo systemctl start elasticsearch
deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/oss-7.x/apt stable main
Hit:1 http://sn.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://sn.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://sn.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 https://artifacts.elastic.co/packages/oss-7.x/apt stable InRelease
Hit:6 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

```

Vérification

```

root@ababacar-Graylog:/home/ababacar# curl -X GET http://localhost:9200
{
  "name" : "ababacar-Graylog",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "t71F4AyyRRSfiXx15RQyg",
  "version" : {
    "number" : "7.10.2",
    "build_flavor" : "oss",
    "build_type" : "deb",
    "build_hash" : "747e1cc71def077253878a59143cf785afa92b9",
    "build_date" : "2021-01-13T00:42:12.435326Z",
    "build_snapshot" : false,
    "lucene_version" : "8.7.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
root@ababacar-Graylog:/home/ababacar#

```

Graylog

```

root@ababacar-Graylog:/home/ababacar# wget https://packages.graylog2.org/repo/packages/graylog-5.2-repository_latest.deb
sudo dpkg -i graylog-5.2-repository_latest.deb
sudo apt update
sudo apt install -y graylog-server
--2025-06-24 13:54:01-- https://packages.graylog2.org/repo/packages/graylog-5.2-repository_latest.deb
Resolving packages.graylog2.org (packages.graylog2.org)... 2606:4700:3030::6815:4001, 2606:4700:3030::6815:7001, 2606:4700:3030::6815:3001, ...
Connecting to packages.graylog2.org (packages.graylog2.org)|2606:4700:3030::6815:4001|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://graylog-package-repository.s3.dualstack.eu-west-1.amazonaws.com/packages/graylog-5.2-repository_latest.deb?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=20250624T135402Z&X-Amz-SignedHeaders=host&X-Amz-Expires=600&X-Amz-Credential=AKIAIJSI6MCSXPXFVDPIA%2F20250624%2Feu-west-1%2Fs3%2Faws4_request&X-Amz-Signature=96bb4fe702b594a959b2682fca108f507846dd78bd7de3fc36766d89d7b7a8d1 [followi
ng]
--2025-06-24 13:54:02-- https://graylog-package-repository.s3.dualstack.eu-west-1.amazonaws.com/packages/graylog-5.2-repository_late
st.deb?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=20250624T135402Z&X-Amz-SignedHeaders=host&X-Amz-Expires=600&X-Amz-Credential=AKIAIJSI6MCSXPXFVDPIA%2F20250624%2Feu-west-1%2Fs3%2Faws4_request&X-Amz-Signature=96bb4fe702b594a959b2682fca108f507846dd78bd7de3fc36766d89d7b7a8d1
Resolving graylog-package-repository.s3.dualstack.eu-west-1.amazonaws.com (graylog-package-repository.s3.dualstack.eu-west-1.amazonaws
s.com)... 2a05:d050:8020:611:34da:240b::, 2a05:d050:80df:9bc9:345c:36a::, 2a05:d050:80e0:9440:34da:1a40::, ...
Connecting to graylog-package-repository.s3.dualstack.eu-west-1.amazonaws.com (graylog-package-repository.s3.dualstack.eu-west-1.amaz
onaws.com)|2a05:d050:8020:611:34da:240b::|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2088 (2.0K) [application/x-debian-package]
Saving to: 'graylog-5.2-repository_latest.deb'

graylog-5.2-repository_latest.deb 100%[=====] 2,04K --.-KB/s in 0s

```

Mise en service

```

root@ababacar-Graylog:/home/ababacar# sudo systemctl enable graylog-server.service
Synchronizing state of graylog-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable graylog-server
Created symlink /etc/systemd/system/multi-user.target.wants/graylog-server.service → /lib/systemd/system/graylog-server.service.
root@ababacar-Graylog:/home/ababacar# sudo systemctl start graylog-server.service
root@ababacar-Graylog:/home/ababacar#

```

Configuration du mot de passe admin

```
root@ababacar-Graylog:/home/ababacar# pwgen -N 1 -s 96
2fybCzwDVvLWuXq0dSNN0d0LOxFQWtU0EHQD4ThJZjxGAWakn3dC9PmHWiN6ZjWGsXSPcyFghqJe3MPyM3iy4i6cPhhJeu4
root@ababacar-Graylog:/home/ababacar#
```

```
root@ababacar-Graylog:/home/ababacar# echo -n 2fybCzwDVvLWuXq0dSNN0d0LOxFQWtU0EHQD4ThJZjxGAWakn3dC9PmHWiN6ZjWGsXSPcyFghqJe3MPyM3iy4i6cPhhJeu4 | sha256sum
90c180fae7d7212e88a9e7cef0e48259ca8acc6248d55741da3fef564539cf97 -
root@ababacar-Graylog:/home/ababacar#
```

Ajouter le hash dans `/etc/graylog/server/server.conf` à root_password_sha2`.`

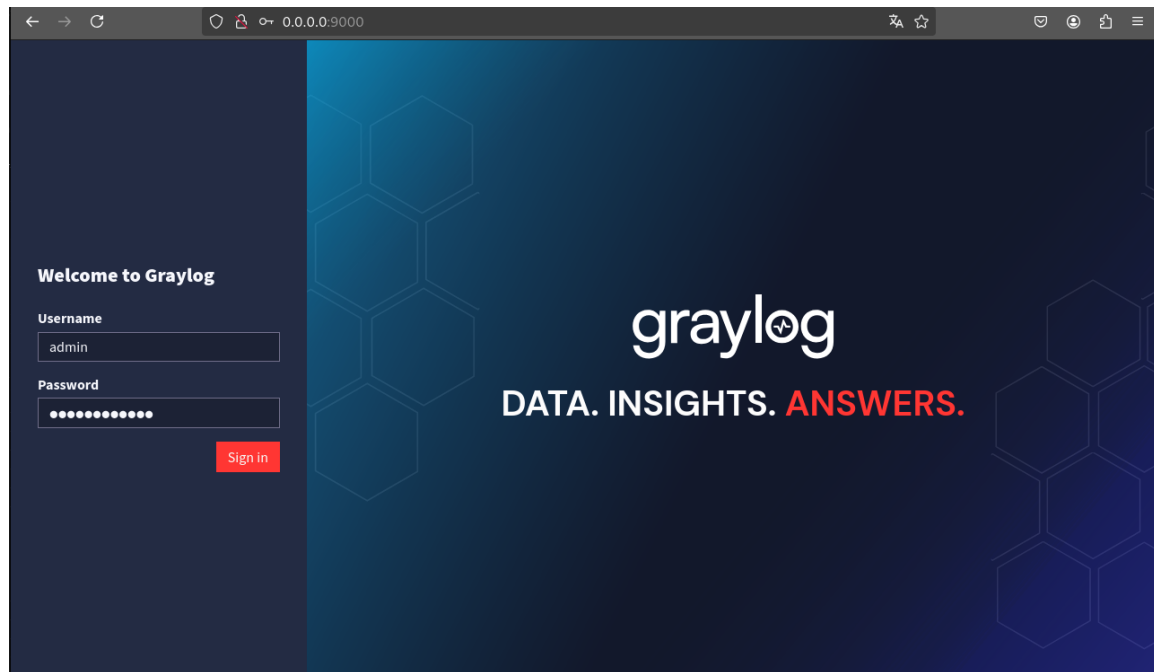
```
password_secret = 2fybCzwDVvLWuXq0dSNN0d0LOxFQWtU0EHQD4ThJZjxGAWakn3dC9PmHWiN6ZjWGsXSPcyFghqJe3MPyM3iy4i6cPhhJeu4
# The default root user is named 'admin'
#root_username = admin

# You MUST specify a hash password for the root user (which you only need to initially set up the
# system and in case you lose connectivity to your authentication backend)
# This password cannot be changed using the API or via the web interface. If you need to change it,
# modify it in this file.
# Create one by using for example: echo -n yourpassword | shasum -a 256
# and put the resulting hash value into the following line
root_password_sha2 = sha256sum
```

Démarrage et vérification

```
sudo systemctl enable --now graylog-server
```

Accès à l'interface: <http://0.0.0.0:9000>



Accès via navigateur à l'interface Web de Graylog.

The screenshot shows the Graylog web interface. The top navigation bar includes links for Search, Streams, Alerts, Dashboards, Enterprise, Security, and System / Users and Teams. The main content area is titled 'Users Overview' and shows an overview of registered users. Below this, there is a section for the 'System Administrator' user, which is locked for editing. A table lists the system administrator user. At the bottom, there is a section for 'Users' with a search bar and a table listing registered users.

Full name	Username	E-Mail Address	Client Address	Enabled	Role	Actions
Administrator	admin		127.0.0.1	✓	Admin	System user Edit tokens

Full name	Username	E-Mail Address	Client Address	Enabled	Role	Actions
Sidecar System User (built-in)	graylog-sidecar	sidecar@graylog.local		✓	Reader Sidecar System (Internal)	Edit More

Ajout d'un utilisateur

The screenshot shows the Graylog web interface after adding a new user. The 'Users' section now displays two registered users in the table.

Full name	Username	E-Mail Address	Client Address	Enabled	Role	Actions
Ababacar Niang	Bomboclat	ababacarniang98@gmail.com		✓	Reader Admin	Edit More
Sidecar System User (built-in)	graylog-sidecar	sidecar@graylog.local		✓	Reader Sidecar System (Internal)	Edit More

Conclusion

La plateforme Graylog est désormais opérationnelle. Cette installation permet de centraliser les logs pour une analyse efficace, dans le cadre de la supervision système ou réseau.
